Computer Algorithms

Recurrence Relations

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Recurrence To Runtime

$$T(n) = T(n/2) + O(1)$$

$$\Big\downarrow$$

$$T(n) = O(\log n)$$

Recurrence To Runtime

$$T(n) = 4T(n/2) + O(n)$$

$$\downarrow$$

$$T(n) = O(n^2)$$

Recurrence To Runtime

$$T(n) = 3T(^n/_2) + O(n)$$

$$\downarrow$$

$$T(n) = O(n^{\log_2 3})$$

Recurrence To Runtime

$$T(n) = 2T(n/2) + O(n)$$

$$\downarrow$$

$$T(n) = O(n \log n)$$

Master Theorem

Theorem

If $T(n) = aT\left(\left\lceil \frac{n}{b} \right\rceil\right) + O(n^d)$ (for constants $a > 0, b > 1, d \geq 0$), then:

$$T(n) = \begin{cases} O(n^d) & \text{if } d > \log_b a \\ O(n^d \log n) & \text{if } d = \log_b a \\ O(n^{\log_b a}) & \text{if } d < \log_b a \end{cases}$$

Master Theorem Example 1

Master Theorem Example 1

$$T(n) = 4T\left(\frac{n}{2}\right) + O(n)$$

$$a = 4$$

$$b = 2$$

$$d = 1$$

Since $d < \log_b a$, $T(n) = O(n^{\log_b a}) = O(n^2)$

Master Theorem Example 2

Master Theorem Example 2

$$T(n) = 3T\left(\frac{n}{2}\right) + O(n)$$

$$a = 3$$

$$b = 2$$

$$d = 1$$

Since $d < \log_b a$,
$T(n) = O(n^{\log_b a}) = O(n^{\log_2 3})$

Master Theorem Example 3

Master Theorem Example 3

$$T(n) = 2T\left(\frac{n}{2}\right) + O(n)$$

$$a = 2$$

$$b = 2$$

$$d = 1$$

Since $d = \log_b a$,

$T(n) = O(n^d \log n) = O(n \log n)$

Master Theorem Example 4

Master Theorem Example 4

$$T(n) = T\left(\frac{n}{2}\right) + O(1)$$

$$a = 1$$

$$b = 2$$

$$d = 0$$

Since $d = \log_b a$, $T(n) = O(n^d \log n) = O(n^0 \log n) = O(\log n)$

Master Theorem Example 5

Master Theorem Example 5

$$T(n) = 2T\left(\frac{n}{2}\right) + O(n^2)$$

$$a = 2$$

$$b = 2$$

$$d = 2$$

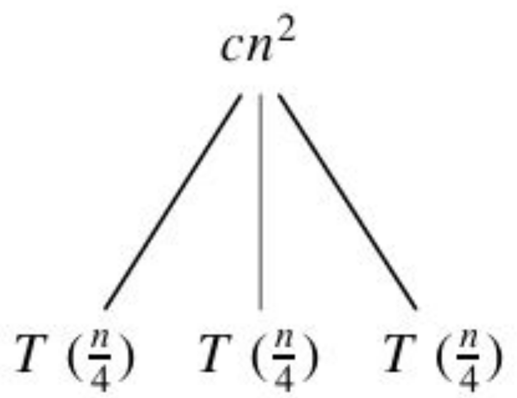
Since $d > \log_b a$, $T(n) = O(n^d) = O(n^2)$

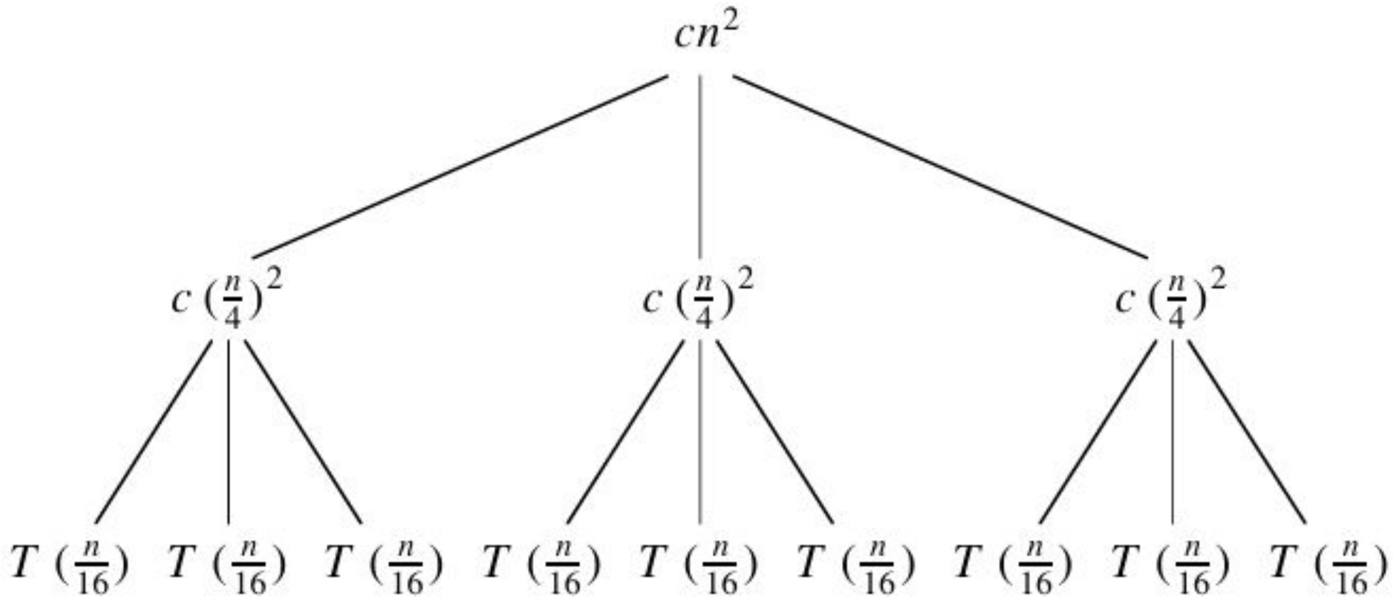
Recursion Tree Method

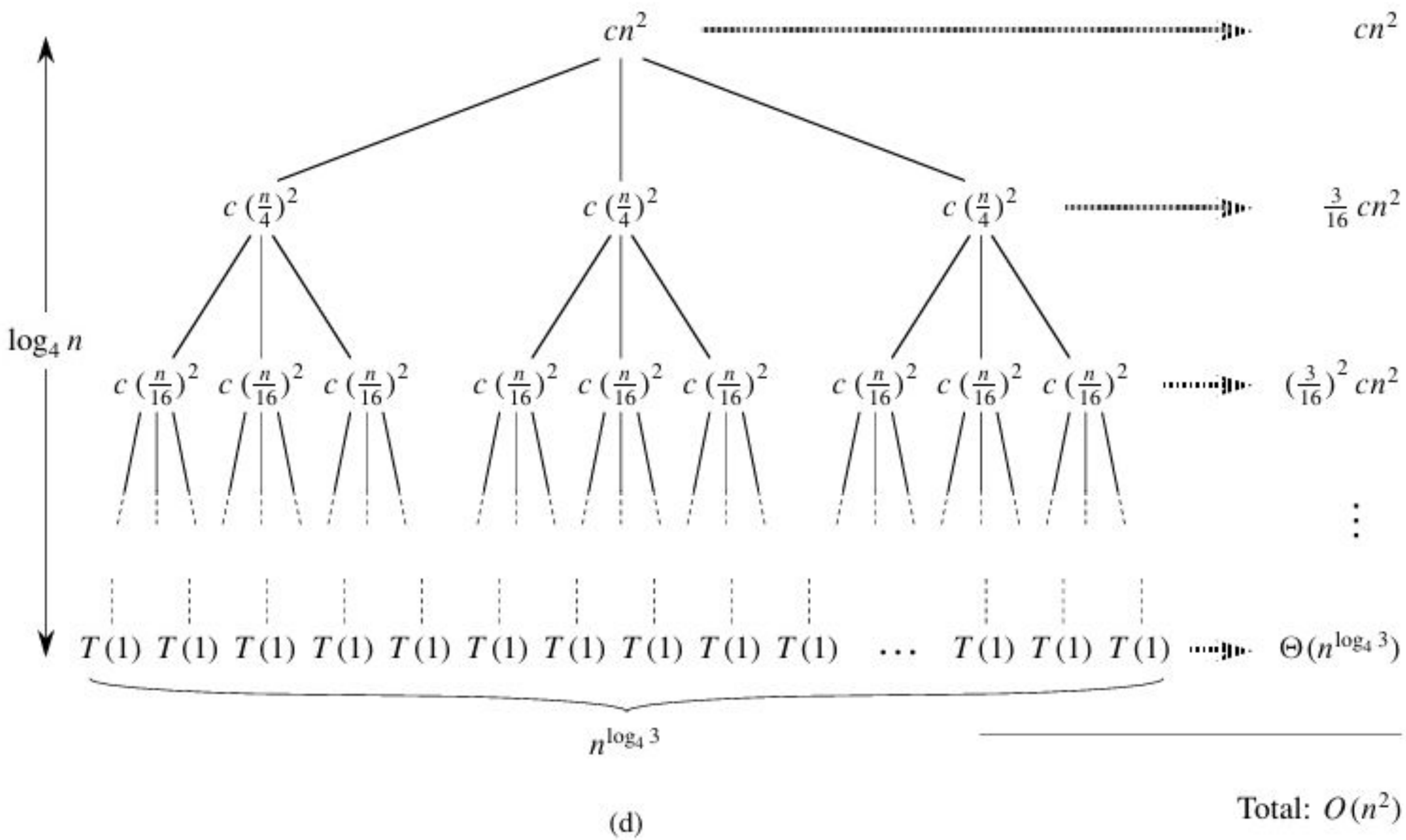
A recursion tree is useful for visualizing what happens when a recurrence is iterated. It diagrams the tree of recrusive calls and the amount of work done at each call.

$$T(n) = 3T\left(\frac{n}{4}\right) + \theta(n^2)$$

Recursion Tree Method







cn^2
$c\left(\frac{n}{4}\right)^2$
$c\left(\frac{n}{16}\right)^2$
$\log_4 n$
$T(1)$
$n^{\log_4 3}$
cn^2
$\frac{3}{16}cn^2$
$\left(\frac{3}{16}\right)^2 cn^2$
$\Theta(n^{\log_4 3})$
Total: $O(n^2)$

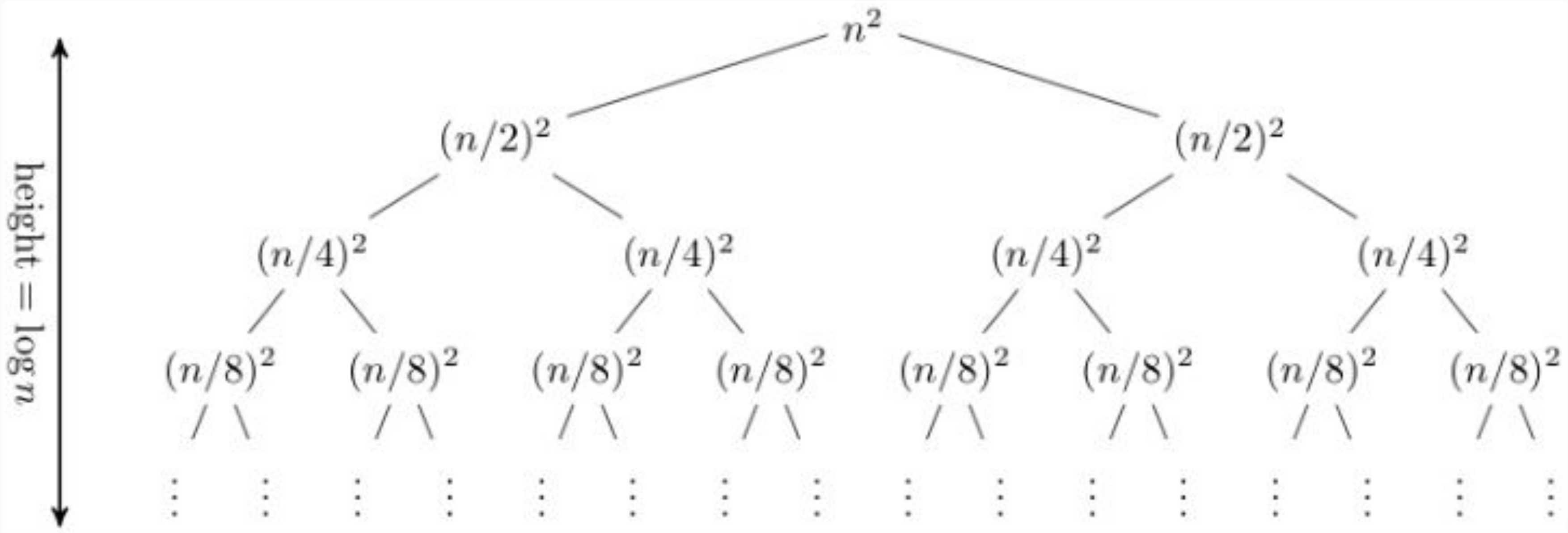
(d)

Solving the equation

$$
\begin{aligned}
T(n) &= \sum_{i=0}^{\log_4 n-1} \left(\frac{3}{16}\right)^i cn^2 + \Theta(n^{\log_4 3}) \\
&< \sum_{i=0}^{\infty} \left(\frac{3}{16}\right)^i cn^2 + \Theta(n^{\log_4 3}) \\
&= \frac{1}{1-(3/16)} cn^2 + \Theta(n^{\log_4 3}) \\
&= \frac{16}{13} cn^2 + \Theta(n^{\log_4 3}) \\
&= O(n^2) .
\end{aligned}
$$

$$T(n) = 2T\left(\frac{n}{2}\right) + n^2$$

The recursion tree for this recurrence has the following form:



References